

Taemin Kim

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EDUCATION

University of Michigan

Ann Arbor, MI

Bachelor of Science in Computer Science and Data Science, Minor in Mathematics & Spanish

May 2027

GPA: 3.77 / 4.00

Coursework: Web Systems, Distributed Systems, Data Structures and Algorithms, Practical Data Science,
Mobile App Development, Machine Learning, Computer Vision, Adv. Software Design

WORK EXPERIENCE

Center for Naval Research and Education

Ann Arbor, MI

Undergraduate Research Assistant

May 2026 - Present

- Conducting research on hybrid deep-learning methods to estimate solid stress and material properties from surrounding flow fields, using Python/JAX implementations of physics-informed models including PINNs, FNOs, and U-Nets
- Created visualizations of predicted flow and stress fields across time using Matplotlib animations
- Experimenting with physics-informed loss functions, model depth, and U-Fourier layer configurations to compare accuracy and generalization across neural operator architectures

LEADERSHIP EXPERIENCE

Michigan EcoData

Aug 2024 - Present

Lead Developer

- Co-led 8-person team collaborating with Office of Campus Sustainability, Radial Power, and Students for Clean Energy on education modules for multi-phase solar rollout and integrated Agile workflows using Jira and GitHub
- Coordinated technical workshops and compiled tutorial documentation to onboard 15+ new contributors to organization

Circadian (Mobile App Development Capstone)

Aug 2025 – Dec 2025

- Led development of iOS sleep tracking app in 6-member Agile team, combining Swift frontend and Docker Compose-managed Go backend and PostgreSQL services, with shared design docs and API specifications to coordinate frontend and backend sub teams
- Implemented account management, authentication, onboarding flow, and friend-network neighborhoods with gamified features like streaks, leaderboards, and friend reminders to encourage accountability
- Integrated Apple HealthKit and Google/Apple Calendar APIs to collect sleep data and generate AI-assisted sleep schedule suggestions via LLMs

PROJECTS

Intra EU Migration Dashboard (Data Visualization)

June 2025 - Present

- Built interactive dashboard visualizing EU migration flows (2004-2023) with time-series animation, country drill-downs, top inflow/outflow rankings, and total EU data using React, TypeScript, Tailwind, and deck.gl
- Cleaned Eurostat TSVs and published three datasets (total bilateral flows, age-group flows, net bilateral flows) using Python, pandas, and NumPy; normalized ISO codes, filtered dimensions and computed net flows
- Motivated by curiosity about how EU free movement has shaped demographic and economic shifts, designed dashboard to explore and share migration patterns in more accessible way

Distributed Search Engine

Apr 2024

- Built scalable search engine from scratch using Hadoop-style MapReduce framework, implementing Manager-Worker system with TCP/UDP sockets, JSON protocols, and heartbeat-based fault tolerance
- Designed 7-stage MapReduce pipeline (mappers/reducers modeled on Hadoop) to construct inverted index from Wikipedia crawls, incorporating text cleaning, tf-idf weighting, cosine similarity, and PageRank integration
- Developed RESTful Index Server to expose inverted index segments and Flask-based Search Server that aggregates results across shards with parallel API calls and ranked retrieval

SKILLS

Programming Languages: Python, JavaScript/TypeScript, SQL (Postgres), C++, C#, C, Golang, Java

Technologies: Git, Shell Scripting (bash), HTML, CSS (Tailwind), React, Hadoop, Flask, Prisma, REST API, AWS,
Agile (Jira/Scrum), Docker, Spring Boot

Data Science Libraries: pandas, NumPy, Plotly, scikit-learn, PyTorch, matplotlib, JAX

Languages: English (Native), Korean (Native), Spanish (Proficient)